

Sea-level rise causes shorebird population collapse before habitats drown

Received: 11 October 2023

Accepted: 29 May 2024

Published online: 03 July 2024



Martijn van de Pol^{1,2}✉, Liam D. Bailey³, Magali Frauendorf^{2,4},
Andrew M. Allen^{1,2,5,6}, Martijn van der Sluijs², Nadia Hijner^{1,2,7},
Lyanne Brouwer¹, Hans de Kroon⁵, Eelke Jongejans^{1,2,5} & Bruno J. Ens^{1,8,9}

Sea-level rise will lead to widespread habitat loss if warming exceeds 2 °C, threatening coastal wildlife globally. Reductions in coastal habitat quality are also expected but their impact and timing are unclear. Here we combine four decades of field data with models of sea-level rise, coastal geomorphology, adaptive behaviour and population dynamics to show that habitat quality is already declining for shorebirds due to increased nest flooding. Consequently, shorebird population collapses are projected well before their habitat drowns in this UNESCO World Heritage Area. The existing focus on habitat loss thus severely underestimates biodiversity impacts of sea-level rise. Shorebirds will also suffer much sooner than previously thought, despite adapting by moving to higher grounds and even if global warming is kept below 2 °C. Such unavoidable and imminent biodiversity impacts imply that mitigation is now urgently needed to boost the resilience of marshes or provide flood-safe habitat elsewhere.

Coastal zones provide vital ecological functions as foraging habitats, migration stopover sites and reproductive habitats for many plant, insect, fish, mammal and bird species. However, intertidal flats, mangroves, beaches and tidal marshes are globally under threat by accelerating sea-level rise (SLR) caused by anthropogenic global warming^{1,2}. Coastal habitats can cope with some SLR through vertical growth (sediment accretion) and inland migration (for example, forest on higher grounds changing into marsh)^{3–8}. However, both modelling and empirical palaeo and contemporary studies show that coastal habitats have a critical rate of SLR, beyond which vertical growth can no longer keep pace and habitat will eventually drown and change into intertidal flats or sea^{1,3,5–7}. SLR up to the year 2100 may exceed these critical rates in many of the coastal zones of the world if warming exceeds 2 °C (ref. 1). If inland migration is impossible—as a result of sea walls, steep topography or the absence of hinterland on small islands—then this ultimately leads to habitat loss^{3,5–8}.

Biodiversity impact assessments of SLR use geomorphological accretion models to quantify the timing and amount of habitat loss that can be expected under different scenarios of future SLR^{9–11}. Habitat loss is subsequently translated into biodiversity loss by multiplying the amount of habitat lost with the current population density of species living in affected areas^{12–17}. As coastal habitat loss is mainly predicted to occur later this century, when SLR further accelerates beyond critical thresholds and historical elevation capital becomes exhausted^{1,10}, most impacts on biodiversity are predicted to also happen many decades from now^{12–17}. Consequently, SLR is not always on the action horizon of managers of coastal wildlife, as habitat loss is still distant, uncertain and possibly preventable (limiting global warming below 2 °C this century can still avert widespread marsh loss due to SLR¹).

However, coastal biodiversity already suffers from SLR, even when habitat loss has not yet occurred. For example, coastal plants and forest performance decline under higher inundation frequency and salt

¹College of Science and Engineering, James Cook University, Townsville, Queensland, Australia. ²Department of Animal Ecology, Netherlands Institute for Ecology (NIOO-KNAW), Wageningen, the Netherlands. ³Leibniz Institute for Zoo and Wildlife Research, Berlin, Germany. ⁴Department of Wildlife, Fish, and Environmental Studies, Swedish University of Agricultural Sciences, Umeå, Sweden. ⁵Department of Ecology, RIBES, Radboud University, Nijmegen, the Netherlands. ⁶Department of Animal Husbandry, Van Hall Larenstein University of Applied Sciences, Velp, the Netherlands. ⁷Groningen Institute for Evolutionary Life Sciences (GELIFES), University of Groningen, Groningen, the Netherlands. ⁸Sovon Dutch Centre for Field Ornithology, Den Burg, the Netherlands. ⁹Royal Netherlands Institute for Sea Research (NIOZ), Den Burg, the Netherlands. ✉e-mail: martijn.vandepol@jcu.edu.au

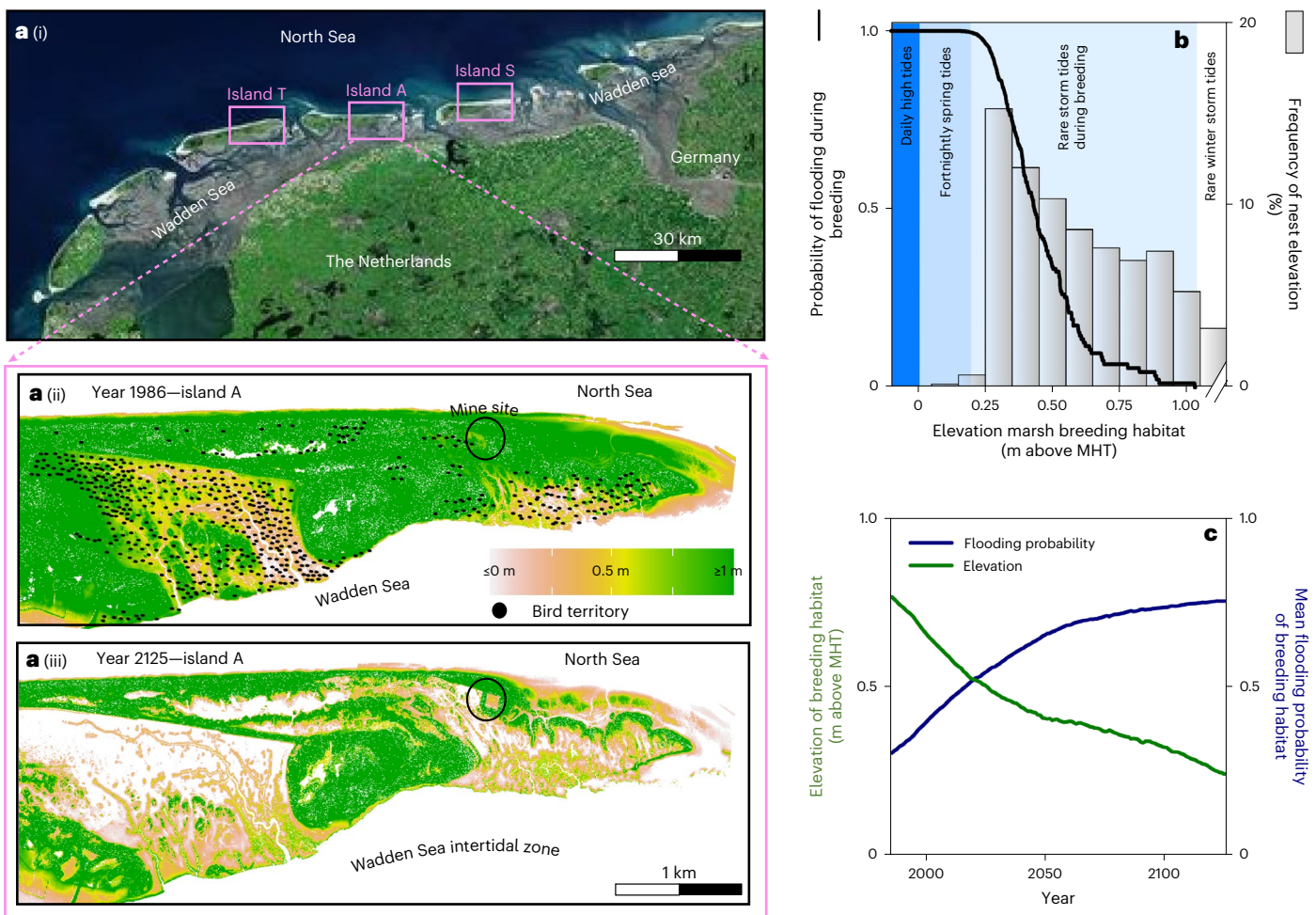


Fig. 1 | Relationship between the elevation of shorebird habitat and flooding risk under SLR. a, Map of study islands Terschelling (T), Ameland (A) and Schiermonnikoog (S) in the western Wadden Sea (i); distribution of oystercatchers breeding territories on an elevation map for 1986 (ii) and a projected marsh elevation map of island A in 2125 (scenario RCP 4.5 and gas mining) (iii). **b**, The relationship between the elevation of the breeding habitat

on the tidal marsh and the probability that it will flood at least once during the breeding season (based on water levels of 1986–2021) (black line) and how this compares to the distribution of nest elevations (grey shaded bars). **c**, Projected changes over time in median elevation and the probability that breeding habitat floods on island A.

intrusion caused by SLR^{18,19}. Shorebirds can forage less if the tidal flats where they feed are exposed for shorter periods²⁰ and they can raise fewer offspring if their nests are flooded more often⁴. Reduction in habitat quality is rarely incorporated into impact assessments of SLR, implying that the current focus on habitat loss underestimates biodiversity impacts: population loss may happen sooner than expected even if global warming, SLR and habitat loss are limited.

Effects on habitat quality are rarely included as it is typically unclear how they translate into changes in birth or death rates and ultimately population size^{21–24}: this often depends on the behaviour of organisms, which requires comprehensive study. The absence of behavioural and demographic mechanisms in most SLR impact studies also means that they ignore adaptive responses of wildlife (for example, by selecting higher nesting sites which are less prone to flooding^{25,26}).

Here we study how SLR affects marsh-nesting shorebirds by combining nearly four decades of behavioural and demographic data with a suite of coupled abiotic and biotic dynamical models (Extended Data Fig. 1 and Extended Data Table 1). We quantify the impact of various GHG emission scenarios of the Intergovernmental Panel on Climate Change (IPCC) and associated rates of SLR on three Dutch island populations of Eurasian oystercatchers (*Haematopus ostralegus*) in the Wadden Sea, the largest intertidal system in the world. Mining on one

island has resulted in up to 0.5 m of deep soil subsidence since 1986, causing relative SLR at a rate so fast that we otherwise would only have been able to study in the future under extreme emission scenarios. Mining thus provided a unique opportunity to empirically quantify key responses to future SLR. We model the mechanism by which flooding events impact nesting success and habitat quality and account for the ability of birds to adapt their nest and habitat selection in response to nest flooding. By quantifying the timing and extent of population and habitat loss we determine the impact of SLR, how SLR affects habitat quality and by how much a focus on habitat loss per se underestimates biodiversity impacts in this UNESCO World Heritage Area.

How SLR affects the flooding of bird nests

On the islands Terschelling, Ameland and Schiermonnikoog (henceforth T, A and S; map Fig. 1a(i)), 78–91% of oystercatchers historically bred on marsh 0–1 m above mean high tide (MHT; all reported land elevations are relative to the sea level of the year considered). Oystercatchers prefer nesting on the lower marsh owing to short nesting vegetation and proximity to intertidal feeding grounds²⁵ and consequently have rarely bred in widely available higher elevation areas (Fig. 1a(ii) and Extended Data Fig. 2 for other islands). The zone of 0–1 m above MHT encompasses virtually the entire range of 0–100% flooding risk in

the breeding season (Fig. 1b): elevations below MHT are flooded daily during normal high tides, elevations 0–0.2 m above MHT are typically flooded every 2 weeks during spring tides (oystercatchers incubate eggs for 4 weeks), while nests between 0.2 and 1 m above MHT are only flooded during storms in the nesting season. Marsh ≥ 1 m above MHT is generally safe during the nesting season but sediment can accrete there during severe winter storms.

Marshes lose elevation relative to the sea level when rates of SLR exceed vertical growth, ultimately leading to habitat loss once elevation falls below MHT and marsh transitions into intertidal flat (white areas in Fig. 1a(iii) that were non-white in Fig. 1a(ii); Extended Data Fig. 2). However, before that happens, small changes in marsh elevation already cause large changes in the flooding probability of nests during the breeding season. For example, a 5 cm elevation loss increases the flooding probability by 17% (from 58% to 75%) for a breeding territory situated 0.4 m above MHT (Fig. 1b), illustrating that every centimetre can matter for nesting shorebirds. For island A, we project a median decrease of 0.5 m in elevation relative to sea level by 2125, which corresponds to a 45% increase in mean flooding probability for the nesting population (Fig. 1c), demonstrating how strongly SLR can erode habitat quality over time.

Birds suffer much earlier from SLR than their habitat drowns

Global analyses have suggested that, in general, tidal marshes are likely to drown ultimately when the rate of SLR is 3–4 mm yr⁻¹ (which is roughly the current rate of SLR²⁷), while they are very likely to drown when SLR exceeds 7–8 mm yr⁻¹ (ref. 1). However, the specific local critical rate of SLR which leads to habitat loss depends on the vertical accretion rate, which in turn depends on the elevation, compaction, distance to the coastline, sediment concentration, tidal amplitude and flooding attenuation^{5,28,29} of a site. Thus, critical rates of SLR will vary both within and among islands. Using a geomorphological accretion model parameterized with long-term field data from our study islands²⁸, we integrated critical rates over the marsh of each island by calculating the rate of SLR beyond which most area used by oystercatchers will experience relative elevation loss (and thus ultimately drown).

The critical rate of SLR was twice as high for some islands compared to others (2.8–5.6 mm yr⁻¹). The current rate of SLR already exceeds the critical threshold on island A (Fig. 2a), consistent with field measurements showing that only this island has been losing elevation relative to sea level²⁸. Importantly, birds already suffer at much lower rates of SLR in terms of population growth because the flooding risk of nests increases for any additional SLR beyond 0 mm yr⁻¹ (Fig. 2b,c). The critical rate of SLR for birds is thus exceeded well before the critical rate for marshes is.

Why bird populations suffer even when the critical rate of marshes is not exceeded can be understood by considering a situation with a low rate of SLR (0–2 mm yr⁻¹). In such a situation, most—but not all—breeding habitat gains elevation relative to sea level and experiences decreasing nest flooding risk over time. However, for the bird population, these gains are outweighed by the historically most productive low sites experiencing strongly increased flooding risk and reduced reproductive success. Oystercatchers prefer to nest in short vegetation near the shoreline close to their food²⁵ but this productive zone on the low marsh is extremely sensitive to SLR (that is, flooding risk is most sensitive to elevation change in this zone -0.25–0.5 m above MHT; Fig. 1c). This also highlights that subtle changes in habitat quality can be crucial.

Impacts of future emission and SLR scenarios

To project future population and habitat loss, we considered three GHG emission scenarios from the latest IPCC models³⁰ downscaled to the Netherlands in terms of SLR²⁷: a low (representative concentration pathway (RCP) 2.6; <2 °C warming), intermediate (RCP 4.5;

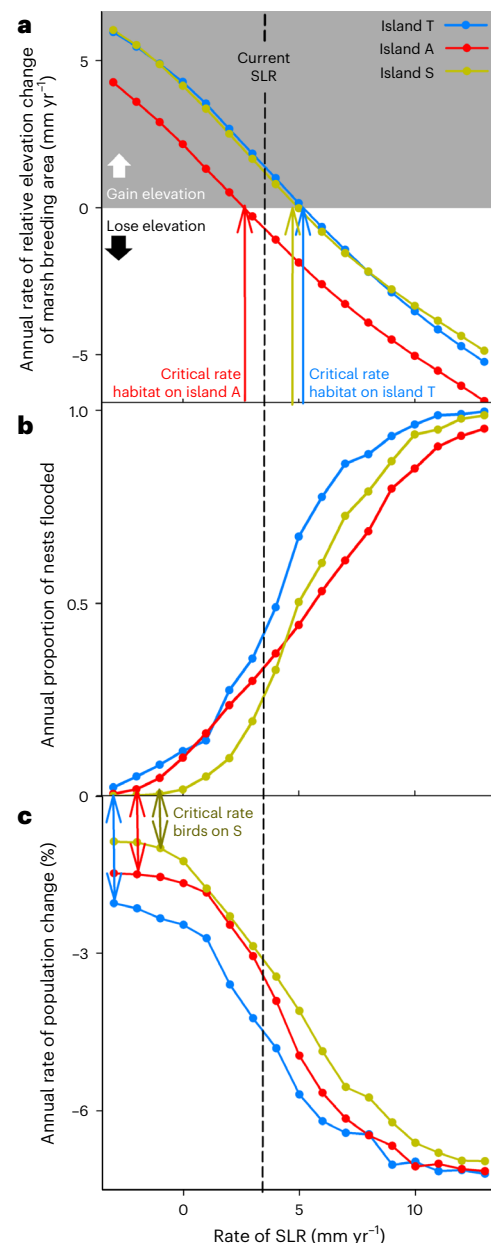


Fig. 2 | Differences in the critical rates of SLR that habitat and birds can withstand on islands T, A and S. a–c. Shown is the impact of SLR rate on annual rate of relative marsh elevation change (a), nest flooding risk (b) and population growth rate (if SLR rate persists for a long time; 200 yr) (c). In a, a negative rate of elevation change implies that most of the marsh is losing elevation relative to sea level and thus the points where lines cross zero give the critical rate of SLR for habitat on each island (arrows). In b and c, the flooding risk and population growth rate already start to deteriorate at much lower values of SLR (arrows) and thus birds have much lower critical rates of SLR (arrows).

<3 °C warming) and very high (RCP 8.5; >4 °C warming) scenario, for which the rate of SLR peaks at different levels and times in the future (Fig. 3a). To quantify the impact of anthropogenic SLR, we compared the above scenarios to that of a hypothetical null-scenario with no further emissions in which the rate of SLR does not accelerate and remains at the average level of twentieth century (no SLR acceleration (SLRA); 1.7 mm yr⁻¹; ref. 27).

A lower critical threshold rate of SLR for birds than for their habitat (previous section) results in large population loss (20–50% among the islands under RCP 4.5) before any habitat loss occurs (arrows Fig. 3b;

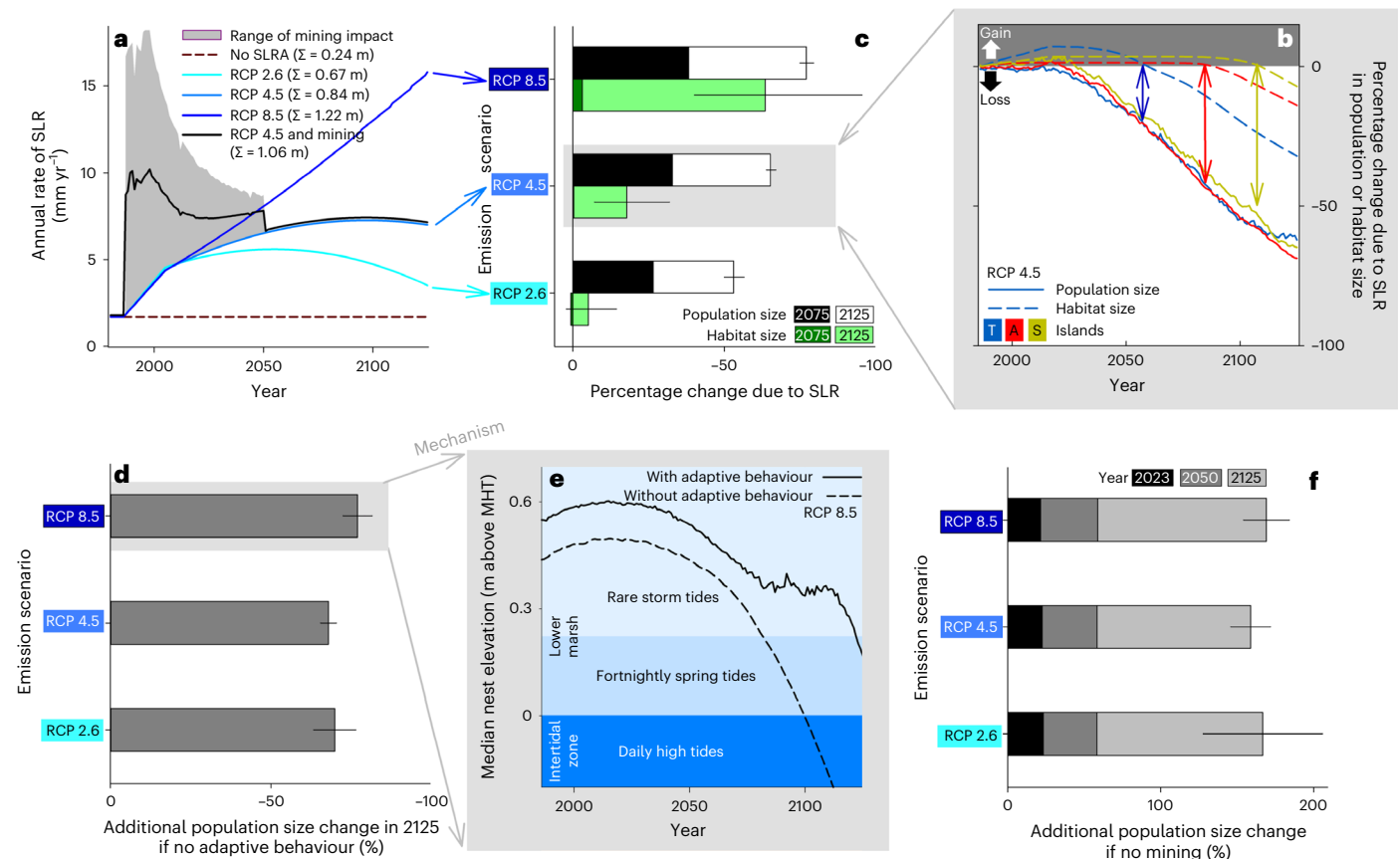


Fig. 3 | The impact of SLR on habitat and oystercatcher population loss for different GHG emission and mining scenarios. **a**, Changes over time in rates of SLR for emission scenarios (Σ gives the summed SLR for 1986–2125). **b**, Changes in population and habitat size over time for each of the islands (RCP 4.5), with arrows showing the population loss at the point where the first net habitat loss occurs. **c**, Habitat and population loss in 2075 and 2125 for each emission scenario (averaged across islands). **d**, Further population size change (decrease) if birds do not behaviourally adapt their nest-site and habitat selection to SLR. **e**, Changes over time in relative nest elevation for a situation with and without adaptive behaviour (island S; RCP 8.5). **f**, Further population size change (increase) if

there had been no gas mining. On island A, subsidence due to gas mining causes further relative SLR from 1986 to 2050; therefore, in **a** the black line and grey area, respectively, show the median additive mining effect and its range across all breeding habitat there. In **b** and **c**, population size change is calculated relative to a reference scenario of no SLRA to determine the additional impact of climate change, while in **d** and **f** we compared a scenario without and with adaption and mining, respectively. In **c** and **d**, error bars depict the range across the means of the three islands and in **f** the range across the means of the low and high estimates of mining subsidence scenarios⁴⁵.

similar results for other emission scenarios in Extended Data Fig. 3). Even once habitat loss starts, population loss exceeds habitat loss (Fig. 3c). Although population loss is often assumed to be proportional to modelled habitat loss, our results indicate that habitat loss can severely underestimate—and even mask decades of—population loss. Thus, habitat loss is an unreliable proxy of population loss (also Extended Data Fig. 4).

Long-term population size losses due to SLR were high and only varied moderately among emission scenarios: 56–79% in 2125 averaged across islands. This suggests that large impacts are inevitable, as they even occur in scenarios where humanity stringently limits greenhouse emissions to keep warming below 2 °C (RCP 2.6). In all scenarios, about half of the population loss occurred before 2075 (Fig. 3c); strong impacts are thus expected already in the next half-century. By contrast, habitat loss varied widely across emission scenarios (6–57% in 2125 averaged across islands) and mainly occurred after 2075 (Fig. 3c). The long-term future of natural habitat on these islands is thus primarily at risk under the very high emission scenario (RCP 8.5), particularly for low-lying islands such as T (with 96% habitat loss in 2125; Extended Data Fig. 3). However, there is large uncertainty within each climatological projection of SLR, meaning that even under an intermediate emission scenario (RCP 4.5) there is still a small risk that most of the habitat will be lost (Extended Data Fig. 5).

Adaptive behaviour buffers some SLR impacts

Previous studies have shown that within their territory oystercatchers do not adjust their nest-site selection after experiencing flooding events but that they do avoid nesting sites that are below the level of spring tides (recognizable by their pioneer vegetation)³¹. Furthermore, young oystercatchers prefer to settle in habitat that successfully produced young in past years, which are increasingly the higher elevation territories that avoid flooding²⁵. On island S, these behavioural mechanisms allowed oystercatchers to increase nest elevation by 5 mm yr⁻¹ from 1995 to 2016, thus far keeping pace with SLR over this period²⁵.

These adaptive nest-site and habitat selection rules together buffered oystercatchers from much—though far from all—of the future impact of SLR, as simulations without such behaviour resulted in 68–76% higher population size loss (Fig. 3d). The population loss due to SLR reported in the previous section would thus have been much higher without such behavioural responses, highlighting the importance of considering adaptation when assessing SLR impacts.

The buffering effect of adaptive behaviour most strongly acts once nest elevations approach the level of spring tides and flooding risk becomes very high (Fig. 3e). At this point the nest-site selection rule to avoid nest sites below the level of spring tides (ref. 31; Fig. 1b) becomes particularly adaptive. However, our model suggests that the buffering effect only lasts temporarily: the decline in nest—but not

habitat—elevation stabilizes due to adaptive nest-site selection for three decades but the decline resumes after 2115 (Fig. 3e) when most habitat is transformed to intertidal flats and birds run out of viable nesting sites to choose from.

Cumulative impact of gas mining

On island A, deep soil subsidence due to gas mining accumulates with emission scenarios as subsidence also results in relative SLR (Fig. 3a). Gas mining started in 1986 and at present >95% of the gas deposit has been extracted but subsidence is expected to continue until 2050 due to delayed geological processes³². The maximum rate of subsidence reached 15 mm yr⁻¹ at the centre of the mining site, which exceeded the rate of SLR due to global warming fivefold in those early years (Fig. 3a). The rate of subsidence decreased with distance from the drilling site and after 2020 emission-driven SLR exceeds rates of mining-driven subsidence at most locations (deep soil subsidence decelerated after the 2000s, while SLR accelerated concurrently; Fig. 3a).

Our model suggests that without subsidence due to mining the population size would currently have been 21–23% higher, while by 2050 and 2125 it would have been 58–59% and 159–169% higher, respectively, across emission scenarios (Fig. 3f). Most of the mining impact on shorebirds thus occurs decades after gas extraction and subsidence have ceased. Mining is projected to have lowered the average habitat elevation on island A by 0.2 m in 2050, which would have otherwise taken another 62 years of SLR under RCP 4.5 (Extended Data Fig. 6). Besides direct population impacts, mining thus also reduces the window of opportunity to mitigate the ultimately more severe consequences of SLR by more than half a century.

Discussion

Our study shows that SLR can lead to an inevitable collapse of coastal birdlife: even a low-emission scenario that limits warming below 2 °C is projected to reduce all three studied oystercatcher populations by more than half. Much population loss will happen before any habitat loss occurs and thus habitat loss is a poor proxy for population loss. The common approach to quantify future habitat loss and then extrapolate this to future population loss using current population densities will probably be far too optimistic^{12–17}, as it assumes that habitat quality is unaffected by SLR. Instead, our study shows that habitat quality is the first thing that is eroded by SLR—here higher flooding risk resulting in lower shorebird reproduction—and we expect this to be the case for many coastal species globally. For example, winter flooding can lead to mass drowning of mammals on barrier islands³³ and the performance of tidal marsh plants is limited by inundation frequency¹⁸. More research on habitat quality effects is needed to determine the broader impact of SLR on coastal biodiversity globally.

Flooding events are a natural part of coastal habitat and the organisms living there have experienced changing sea levels repeatedly in their evolutionary history. Despite being rarely studied, it should thus not be surprising that coastal organisms have already evolved ways of coping with SLR²⁵. Our model suggests that the adaptive behaviour of choosing recently productive (higher elevation) nesting habitat is helping to buffer populations from many of the impacts of SLR. These behavioural adaptations have historically been sufficient to keep pace with SLR²⁵ but are not enough to fully cope with future SLR. This result echoes a widely shared—but rarely evidenced³⁴—concern that organisms cannot keep up with rapid climate change³⁵.

Mining exacerbates future impacts of SLR; consequently, mining impacts can be much delayed. One thus should not primarily rely on monitoring impacts in situ while mining occurs, as done in the assessment of mining on island A which did not detect any substantial impacts on birds over the past 37 years³⁶. Instead, a long-term perspective based on prospective modelling of how threats accumulate is crucial. This is illustrated by our model showing (1) that the largest mining impacts on wildlife only occur after mining stops and (2) that mining reduces

the window of opportunity to act on future SLR by more than half a century. These insights also support recent warning from UNESCO that a planned approval of new mining activities in the Wadden Sea is incompatible with its World Heritage status³⁷.

Management of other threats may also benefit from considering SLR. Offspring production is the bottleneck for oystercatcher population recovery, which is also too low because of food shortages^{38,39}. Actions aimed at increasing food stocks may lead to less nest predation and chick starvation but this will help little if these nests subsequently flood (flooding losses are highest in years with most food³⁸). Mitigation of SLR impacts by reducing competing threats may have limited value and integrative modelling of threats is needed to identify effective actions.

The barrier islands we studied provide habitat for many bird, plant and insect species for which the Wadden Sea World Heritage Area is globally important⁴⁰. In the highest emission scenario, island marshes could disappear almost entirely within a century, meaning that ultimately their whole ecosystem is at risk if humanity continues as usual. Notwithstanding, nature managers often view habitat loss due to future SLR as being beyond the horizon of their current actions. The reason is that habitat drowning is still many decades away, organisms may adapt and the amount of habitat loss may be limited if we keep global warming below 2 °C, meaning there is still much uncertainty about what action is needed and when. However, our study suggests that SLR already erodes habitat quality and that population impacts are imminent, even if warming is kept below 2 °C and despite wildlife adapting. Thus, action is needed now, as the biodiversity impacts of SLR are expected to occur much sooner than previously thought.

Reductions in emissions are critical to any solution but this is beyond the control of local organizations and the time window to prevent the worst impacts of anthropogenic warming is closing³⁰. For the oystercatchers studied here, a large impact already appears inevitable, as even under the optimistic low-emission scenario SLR will halve their numbers in a century. Notwithstanding, opportunities for local mitigation of impacts exist. Mitigation could aim to increase accretion by allowing more sediment from the North Sea to enter marshes by removing dune barriers⁴¹ or by actively trapping sediment on the marsh (using drainage ditches and sedimentation fields surrounded by brushwood groynes)⁴². However, such actions may not be sufficient to safeguard present marsh communities in situ. Nature management may also need to shift focus towards other areas that provide similar ecosystem functions elsewhere, such as artificial lagoons and farmland behind sea walls that provide flood-safe breeding areas⁴³. Oystercatchers have historically been successful in colonizing new habitat⁴⁴. However, many other coastal species are habitat specialists and less flexible, meaning that options to compensate for loss of core habitat are limited. Although it is still unclear how successful mitigation actions could be, it is clear that drastic shifts in thinking about SLR are needed to protect the Wadden Sea and possibly other coastal ecosystems across the globe.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41558-024-02051-w>.

References

1. Saintilan, N. et al. Widespread retreat of coastal habitat is likely at warming levels above 1.5 °C. *Nature* **621**, 112–119 (2023).
2. He, Q. & Silliman, B. R. Climate change, human impacts and coastal ecosystems in the Anthropocene. *Curr. Biol.* **29**, R1021–R1035 (2019).
3. Saintilan, N. et al. Thresholds of mangrove survival under rapid sea level rise. *Science* **368**, 1118–1121 (2020).

4. van de Pol, M. et al. Do changes in the frequency, magnitude and timing of extreme climatic events threaten the population viability of coastal birds? *J. Appl. Ecol.* **47**, 720–730 (2010).
5. Kirwan, M. L., Temmerman, S., Skeehan, E. E., Guntenspergen, G. R. & Fagherazzi, S. Overestimation of marsh vulnerability to sea level rise. *Nat. Clim. Change* **6**, 253–260 (2016).
6. FitzGerald, D. M. & Hughes, Z. Marsh processes and their response to climate change and sea-level rise. *Annu. Rev. Earth Planet. Sci.* **47**, 481–517 (2019).
7. Leuven, J. R. F. W., Pierik, H. J., Vegt, M. van der, Bouma, T. J. & Kleinhans, M. G. Sea-level-rise-induced threats depend on the size of tide-influenced estuaries worldwide. *Nat. Clim. Change* **9**, 986–992 (2019).
8. Pontee, N. Defining coastal squeeze: a discussion. *Ocean Coast. Manag.* **84**, 204–207 (2013).
9. Thorne, K. M., Buffington, K. J., Elliott-Fisk, D. L. & Takekawa, J. Y. Tidal marsh susceptibility to sea-level rise: importance of local-scale models. *J. Fish. Wildl. Manag.* **6**, 290–304 (2015).
10. Crosby, S. C. et al. Salt marsh persistence is threatened by predicted sea-level rise. *Estuar. Coast. Shelf Sci.* **181**, 93–99 (2016).
11. Galbraith, H. et al. Global climate change and sea level rise: potential losses of intertidal habitat for shorebirds. *Waterbirds* **25**, 173–183 (2002).
12. Schmidt, J. A., McCleery, R., Seavey, J. R., Cameron Devitt, S. E. & Schmidt, P. M. Impacts of a half century of sea-level rise and development on an endangered mammal. *Glob. Change Biol.* **18**, 3536–3542 (2012).
13. Leonard, P. B. et al. Landscape connectivity losses due to sea level rise and land use change. *Anim. Conserv.* **20**, 80–90 (2017).
14. Von Holle, B. et al. Effects of future sea level rise on coastal habitat. *J. Wildl. Manag.* **83**, 694–704 (2019).
15. Klingbeil, B. T. et al. High uncertainty over the future of tidal marsh birds under current sea-level rise projections. *Biodivers. Conserv.* **30**, 431–443 (2021).
16. Iwamura, T. et al. Migratory connectivity magnifies the consequences of habitat loss from sea-level rise for shorebird populations. *Proc. R. Soc. B* **280**, 20130325 (2013).
17. Traill, L. W. et al. Managing for change: wetland transitions under sea-level rise and outcomes for threatened species: threatened populations and sea-level rise. *Divers. Distrib.* **17**, 1225–1233 (2011).
18. Pennings, S. C., Grant, M.-B. & Bertness, M. D. Plant zonation in low-latitude salt marshes: disentangling the roles of flooding, salinity and competition. *J. Ecol.* **93**, 159–167 (2005).
19. Saha, A. K. et al. Sea level rise and South Florida coastal forests. *Clim. Change* **107**, 81–108 (2011).
20. van der Kolk, H.-J. et al. Shorebird feeding specialists differ in how environmental conditions alter their foraging time. *Behav. Ecol.* **31**, 371–382 (2020).
21. Stillman, R. A. et al. Predicting effects of environmental change on a migratory herbivore. *Ecosphere* **6**, art114 (2015).
22. Pike, D. A., Roznik, E. A. & Bell, I. Nest inundation from sea-level rise threatens sea turtle population viability. *R. Soc. Open Sci.* **2**, 150127 (2015).
23. Field, C. R. et al. High-resolution tide projections reveal extinction threshold in response to sea-level rise. *Glob. Change Biol.* **23**, 2058–2070 (2017).
24. Roberts, S. G. et al. Preventing local extinctions of tidal marsh endemic Seaside Sparrows and Saltmarsh Sparrows in eastern North America. *Condor* **121**, duy024 (2019).
25. Bailey, L. D. et al. Habitat selection can reduce effects of extreme climatic events in a long-lived shorebird. *J. Anim. Ecol.* **88**, 1474–1485 (2019).
26. Benvenuti, B., Walsh, J., O'Brien, K. M. & Kovach, A. I. Plasticity in nesting adaptations of a tidal marsh endemic bird. *Ecol. Evol.* **8**, 10780–10793 (2018).
27. KNMI Klimaatsignaal'21. KNMI www.knmi.nl/kennis-en-datacentrum/achtergrond/knmi-klimaatsignaal-21 (2021).
28. van Dobben, H. F., de Groot, A. V. & Bakker, J. P. Salt marsh accretion with and without deep soil subsidence as a proxy for sea-level rise. *Estuar. Coasts* **45**, 1562–1582 (2022).
29. Stark, J., Van Oyen, T., Meire, P. & Temmerman, S. Observations of tidal and storm surge attenuation in a large tidal marsh. *Limnol. Oceanogr.* **60**, 1371–1381 (2015).
30. IPCC *Climate Change 2023: Synthesis Report* (eds Core Writing Team, Lee, H. & Romero, J.) (IPCC, 2023).
31. Bailey, L. D. et al. No phenotypic plasticity in nest-site selection in response to extreme flooding events. *Philos. Trans. R. Soc. B* **372**, 20160139 (2017).
32. *Gaswinning Vanaf de Locaties Moddergat, Lauwersoog en Vierhuizen: Resultaten Uitvoering Meet- en Regelcyclus 2019* (NAM, 2020).
33. Schrama, M., Kuijper, D. P. J., Veeneklaas, R. M. & Bakker, J. P. Long-term decline in a salt marsh hare population largely driven by bottom-up factors. *Ecoscience* **22**, 71–82 (2015).
34. Radchuk, V. et al. Adaptive responses of animals to climate change are most likely insufficient. *Nat. Commun.* **10**, 3109 (2019).
35. Parmesan, C. Ecological and evolutionary responses to recent climate change. *Annu. Rev. Ecol. Evol. Syst.* **37**, 637–669 (2006).
36. de Vlas, J. *Samenvatting Monitoring Effecten van Bodemdaling op Ameland-Oost: Evaluatie na 30 Jaar Gaswinning* (Waddenacademie, 2017).
37. *State of Conservation of Properties Inscribed on the World Heritage List* (UNESCO, 2023); <https://whc.unesco.org/archive/2023/whc23-45com-7B.Add2-en.pdf>
38. van de Pol, M. et al. Effects of climate change and variability on population dynamics in a long-lived shorebird. *Ecology* **91**, 1192–1204 (2010).
39. Allen, A. M. et al. The demographic causes of population change vary across four decades in a long-lived shorebird. *Ecology* **103**, e3615 (2022).
40. Esselink, P. et al. in *Wadden Sea Quality Status Report* (eds Kloepper S. et al.) 1–40 (Common Wadden Sea Secretariat, 2017).
41. Engelstad, A. et al. Observations of waves and currents during barrier island inundation. *J. Geophys. Res. Oceans* **122**, 3152–3169 (2017).
42. Marin-Diaz, B. et al. Using salt marshes for coastal protection: effective but hard to get where needed most. *J. Appl. Ecol.* **60**, 1286–1301 (2023).
43. Babcock, M. & Booth, V. *Tern Conservation Best Practice: Habitat Creation and Restoration* (Life, 2020); https://roseatetern.org/uploads/3/5/8/0/35804201/babcock_and_booth_2020_habitat_creation_and_restoration_tern_conservation_best_practice.pdf
44. Goss-Custard, J. *The Oystercatcher: From Individuals to Populations* (Oxford Univ. Press, 1996).
45. Piening, H., van der Veen, W. & van Eijs, R. in *Monitoring Effecten van Bodemdaling op Oost-Ameland* (ed. de Vlas, J.) 9–25 (Waddenacademie, 2017).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2024

Methods

Animal ethics

The collection of long-term bird monitoring data on islands S and A, which was used to parameterize our models, was approved by the Animal Experiments Committee (CCD project AVD25000202010465).

Study system

We studied the impact of SLR on three neighbouring barrier islands in the western (Dutch) part of the Wadden Sea UNESCO World Heritage Area: T, A and S (Fig. 1a(i); between 53° and 54° N and 5° and 7° E). The eastern parts of these islands are nature reserves that primarily consist of tidal marsh, which can naturally flood during storm tides, mostly in winter but on rare occasions also during the breeding season in spring/summer. On these three islands are some of the largest and highest density breeding populations of Eurasian oystercatchers, an indicator shorebird species which is listed as near-threatened by the IUCN and has been declining by 3% annually since the 1990s⁴⁶. During this period of decline, tidal flooding events in the nesting season have become more frequent, which has resulted in the failure of many nests and contributed to low reproductive output for this species^{4,38}. On island A, gas mining from 1986 onwards has caused further deep soil subsidence, reaching a maximum of up to 0.5 m subsidence at the centre of the subsidence area in 2050 (ref. 45).

Oystercatchers are long-lived birds with high site-fidelity (both limited natal and breeding dispersal⁴⁷) and often occupy the same territory for most of their lives⁴⁸, with stable territory borders⁴⁹. Oystercatcher pairs are single-brooded (with limited renesting attempts after failure) and breed from May until July, during which they lay a clutch of one to four eggs in a nest scrape on the ground. Eggs are incubated for 4 weeks, with chicks fledging after 3–5 weeks. Eggs that have been incubated for some time can float a bit but, during tidal flooding, eggs often drift away from the nest and either do not hatch or are preyed upon in the chaos that accompanies flooding events. Furthermore, chicks up to 2 weeks old can drown, suffer from hypothermia or get predated during flooding⁴.

Overview of simulation model

We developed a spatially explicit simulation model to quantify how GHG emission scenarios and resulting SLR determine habitat and population size loss of the three island populations of oystercatchers over the period 1986–2125 (Extended Data Fig. 1 gives an overview). Most of the submodels were parameterized and validated specifically for the barrier islands we study here (Extended Data Table 1). The geomorphological model by ref. 28 was parameterized and validated specifically for our three study islands, while the deep soil subsidence model was developed specifically for island A, where gas mining occurs⁴⁵. The SLR model spatially downscales IPCC scenarios to the situation of the Netherlands²⁷. These abiotic models are initialized with empirical input data on historical marsh elevation (LiDAR) and water levels (tidal gauges) from each of the three study islands. Furthermore, all the biotic models are developed on the basis of previous long-term studies of oystercatchers on the study islands of S (1984–2021) and A (2012–2019) and initial bird numbers and distributions were based on territory mapping on all three islands. The complete model structure was implemented in the R language⁵⁰; the input data files and parameter values used are available on Dryad⁵¹.

We performed four analyses. First, to determine at what critical rate of SLR the habitat will start to lose elevation and birds experience increased flooding risk and reduced population growth, we ran the model at a range of SLR values (–3 to +13 mm yr^{–1}) which we kept constant for an extended period (200 yr). Second, to project the future impacts of anthropogenic SLR, we quantified the amount of habitat and population loss from 1986 to 2125 for SLR trajectories resulting from three different GHG emission scenarios (RCP 2.6, 4.5 and 8.5) and compared it to a reference situation in which there would be no emissions and no SLRA. Third, to determine how much adaptive behaviour buffered impacts of climate change, we compared habitat

and population loss from 1986 to 2125 for a situation with and without birds behaviourally adapting their nest-site and habitat selection to flooding events. Fourth, to determine the impact of gas mining and how it interacts with SLR, we compared habitat and population loss from 1986 to 2125 for a situation with and without gas mining (island A only) for each of the emission scenarios.

In all simulations, we ran 100 replicate runs for each scenario and report the median value across these replicates. Initially, we ran models at the highest spatial resolution available from the elevation map (0.5 × 0.5 m²) but this was extremely memory- and CPU-intensive as each territory consisted of roughly 100,000 grid cells (sites) which required updating in the geomorphological model. Therefore, we downgraded the spatial resolution to speed up simulations and noticed that using a 6 × 6 m² spatial map resolution did not affect the model results and used it in all simulations (bird territories cover tens of thousands of square metres). We next describe each submodel in more detail.

SLR and intra- and interannual variability models

The mean flooding level in year y at a given location l is determined by the ground surface elevation relative to the sea level (Z) and is affected by changes in the (high tide) water level (W) and changes in the ground elevation (G).

$$Z_{y,l} = G_{y,l} - W_{y,l} \quad (1)$$

Water levels increase over time due to SLR and further fluctuate across years due to weather variability and various lunar cycles (particularly the 18.6 yr lunar nodal cycle⁵²). Within years, high-tide water levels also fluctuate as a result of the 29.5 day lunar tidal cycle (causing fortnightly spring tides) and weather variation (depending on wind strength and direction), with flooding risk being lowest during the nesting season in May and June and highest in the stormy winter season⁴.

To backtrack the influence of historical sea-level fluctuations and project the consequences of increasing rates of SLR, a model was developed which generated daily high-tide levels for each day for the years 1986–2125. For historical years, measurements of daily high-tide levels were available (1986–2021) from tidal gauge stations on each of the study islands⁵³. For subsequent years ($y > 2021$), high-tide levels W were generated for each day d and study island s by (1) drawing a random year of historical tidal data H (reflecting year-to-year variation in water levels due to weather variation), on which we superimposed (2) the projected cumulative SLR to year y (R) and (3) the difference in the sea level caused by the 18.6 yr lunar nodal cycle (N)⁵².

$$W_{y,d,s} = H_{y=x,d,s} + R_y + N_y \quad (2)$$

with x being a randomly selected year between 1986 and 2021. This modelling approach assumes that the MHT can change across years but that the among- and within-year variability as well as the seasonal patterns in MHT (temporal autocorrelation due to the moon cycle and fewer high storm tides in May–June) will not change directionally. The assumption that the within-year variability will remain constant in the future potentially is conservative, as the highest tides during the breeding season have been increasing at twice the rate as the MHTs in this region in recent decades⁴. However, at present, it is unknown whether such an increase in within-year variability is a consequence of climate change and how it will develop in the future.

The sixth IPCC report³⁰ described how SLR is projected to change across the globe under different socio-economic GHG emission scenarios, which have been spatially downscaled to the situation of the Netherlands by the Royal Netherlands Meteorological Institute²⁷. We considered three scenarios that reflect a low (RCP 2.6), intermediate (RCP 4.5) and very high (RCP 8.5) emission trajectory. Although RCP 8.5 is historically considered an extreme scenario, recent

studies suggest that RCP 8.5 best tracks historical emissions and anticipated outcomes of current global climate policies until 2050⁵⁴. As the above-published climate projections included among-year variability which we model ourselves specifically, we approximated the projections of the above scenarios by the following polynomial functions (all $R^2 > 0.99$):

$$R_y = \sum_{y=t}^{y=y_0} (\rho_0 + \rho_1(t - y_0) + \rho_2(t - y_0)^2) \quad (3)$$

For scenario-specific parameter values, see ref. 51.

The lunar nodal cycle creates more systematic fluctuations in mean sea level among years and can be modelled as^{55,56}:

$$N_y = \frac{v_1}{2 \sin(2\pi) v_2 (y - v_3)} \quad (4)$$

with the duration of the cycle $v_2 = 18.6$ yr and the shift and amplitude of the cycle at the study sites being $v_1 = 0.0222$ m and $v_3 = 2.4$ yr, respectively³¹. Note that we removed the effect of the lunar nodal cycle before selecting water levels of a random year $H_{y=x,d,s}$ in equation (2).

Flooding model

During flooding events, water levels on the marsh are different from water levels on the intertidal estuaries where tidal gauges are located because of amplification of water in creeks and attenuation of water when travelling over the marsh plain. Others⁵⁷ developed a flooding model for tidal marsh elsewhere in the Netherlands and we added their flooding model to equation (2) to incorporate the local attenuation/amplification of water onto the marsh plain via creeks (A_l) that is specific for each location l but does not vary over time:

$$W_{y,d,s,l} = W_{y,d,s} + A_l \quad (5)$$

with

$$A_l = \alpha_1 C_l - \alpha_2 P_l \quad (6)$$

where C_l is the creek length (m) and P_l is the Euclidian distance between location l and the nearest creek (m), $\alpha_1 = 4.0 \times 10^{-5}$ and $\alpha_2 = 3.5 \times 10^{-4}$. If a site was closer to the coastline than to a creek, then we set $C_l = 0$ and calculated P_l as the Euclidian distance of a location to the nearest coastline point as in ref. 28. We assumed that the location of creeks did not change over time, such that C_l and P_l were constant over time.

Map to initialize elevation dynamics

To initialize our spatially explicit model, we constructed an elevation map for each island in the year 1986. A high-resolution spatial elevation map of the study area (0.5×0.5 m² grid resolution) based on LiDAR measurements from 2014 was available⁵⁸. We calibrated LiDAR elevation measurements with in situ measurements of elevation taken with a DGPS on study island S in the same year ($n = 254$). We considered in situ measurements using DGPS to be most reliable, as these have less measurement error (± 2 cm standard error), while LiDAR can also be affected by vegetation reflecting signals and the filtering out of such vegetation reflection (as well as accounting for other sources of noise) may be imperfect. Simple linear regression of DGPS elevation against LiDAR elevation showed that the LiDAR elevation measurements were on average 0.056 m lower than the in situ measurements, with the discrepancy increasing with elevation (probably because marsh vegetation height covaries with elevation). Therefore, we calculated ground elevation G (in metres above Dutch ordnance level) by adjusting the LiDAR elevation (L) to account for this systematic bias:

$$G = \gamma_0 + \gamma_1 L \quad (7)$$

with $\gamma_0 = 0.0263$ and $\gamma_1 = 1.0196$. To translate our 2014 elevation map to the start year 1986 of our simulations, we back-calculated elevations using the geomorphological model described below.

We generated a map of the coastline and creeks essential for the geomorphological model using the elevation map. We selected all grid cells that inundate almost daily (that is, are 0.03 m below or above MHT), which results in a map of all edges of the coastline and all creeks in the study areas. Next, we used the concaveman algorithm⁵⁹ to calculate the border of our map (coastline of the island), while all points further away from the coastline were considered to be creeks. A visual comparison of our coastline and creek map with satellite maps from Google Earth⁶⁰ suggested that this algorithm works well.

Geomorphological model for sedimentation

Vertical changes in ground level are determined by the balance between accretion due to compensatory sedimentation deposition after flooding on the one hand and soil compaction on the other hand. When marshes flood, sediment in the seawater deposits on the marsh and allows the marsh elevation to accrete, resulting in a negative feedback loop between marsh elevation and flooding frequency: SLR causes more frequent flooding, which results in more accretion, which in turn reduces flooding frequency. The existence of such a feedback loop is also empirically supported by observations that accretion rates are typically higher in areas with higher SLR⁵. The rate of sedimentation depends on a variety of aspects, such as the flooding duration, water column height, sediment concentration and the distance the water has travelled (sediment depletes while travelling over the marsh plain). In addition, accreted sediment (clay layer) compacts due to gravity⁶¹. Finally, changes in water content in the deposited clay layer of marshes can lead to swelling or shrinking, causing reversible change.

Here we used an accretion model that was parameterized and validated using long-term empirical measurements on in situ accretion from our study islands²⁸. This model by ref. 28 is uniquely placed to project into the future, as one of the islands on which it is parameterized has experienced gas mining leading to rates of relative SLR that we are only expected to observe under the most extreme emission scenarios in the future (next section). In contrast to most other sedimentation models used elsewhere in the world, projected responses are thus less reliant on extrapolations towards high rates of SLR. We implemented their best-supported accretion model, which had twice the support of their next best model. We note that their model set that covered 95% of the cumulative model weight included (1) effectively the same predictor variables operationalized in a slightly different way, (2) had similar parameter estimates and (3) resulted in similar projections for ground level changes²⁸. Their best model is described by the recurrence equation:

$$G_{y+1,l} = G_{y,l} + \gamma_1 + \gamma_2 \log(M_{y,l} + 1) + \gamma_3 \log(F_{y,l} + 1) + \gamma_4 e^{\gamma_5 P_l} + \gamma_6 C_l + \gamma_7 D_y B_{y,l} \quad (8)$$

where M is the mean water depth (cm) over year y , F is the mean flooding depth in year y (cm), P and C are the plain distance and creek length (m) also used in equation (6), D is the annual net precipitation surplus compared to the previous year (mm, accounted for evaporation loss) and B is the age of the marsh (years since origin, a proxy for the thickness of the accumulated clay layer which is susceptible to shrinkage). Parameter values are deposited in Dryad⁵¹.

The most pessimistic emission scenarios (RCP 8.5) project a roughly linear decrease in net precipitation from 2020 to 2100 of -1.6 mm yr⁻¹ (ref. 27), which would amount at most to a negligible extra -0.000005 m yr⁻¹ elevation change in the ref. 28 model. Thus, although shrinkage/swelling can cause substantial fluctuations among years (± 0.007 m), under current climate projections it is not expected to

strongly contribute to long-term elevation changes. We henceforth ignored shrinkage/swelling, by setting $D_y = 0$ for all years.

Dynamic accretion models generally predict that marshes can keep up with SLR up to a critical rate that will depend on the characteristics of the local conditions⁵. It was shown by ref. 28 that, according to both empirical data and model predictions, marshes on our study islands have so far kept up with current rates of sea level, except on island A. Their model predicts that higher sites and sites further away from the coastline and creeks will start to sink first, with some locations already sinking at rates of SLR of only 2 mm yr⁻¹ (current rates of SLR are 3–4 mm yr⁻¹; ref. 27). Overall, ref. 28 concluded that these islands can keep up with SLR up to 2100 but this conclusion—which our results show was premature—was not supported by spatially integrating projections over the entire islands, as we do here.

Deep soil subsidence due to gas mining

On island A, the extraction of natural gas has locally resulted in deep soil subsidence since 1986. The amount of subsidence as of 2019 cumulated to 0.39 m at the centre of the mining location, with the amount of subsidence decreasing with distance from the mining location, such that at 8 km distance the subsidence due to gas mining is almost zero (and thus neighbouring islands S and T are unaffected). Gas mining is scheduled to continue until the reservoir is depleted (now ~5% remains) and some lagged subsidence due to historical extraction is expected^{32,45}.

To estimate the historical and future amount of subsidence resulting from gas mining, we used the model projections of a geomorphological model that was specifically developed to quantify the local subsidence at this location and which is calibrated and validated with field measurements on deep soil subsidence⁴⁵. As model estimates come with some uncertainty, we considered their base, low and high scenario, reflecting, respectively, the 50%, 2.5% and 97.5% percentile of all model simulations of subsidence. The deep soil subsidence model predicts the subsidence for each year from the start year 1986 to 2050, at which point it is projected to have a cumulative subsidence of maximally 0.48 m (95% percentile: 0.45–0.50 m subsidence). In 2050, annual deep soil subsidence is typically small (most places ~1 mm yr⁻¹; Fig. 3a) and as no model projections beyond 2050 are available³², we set it to zero after 2050.

Soil subsidence model projections were available at a 250 × 250 m² grid resolution and, to match it with the resolution of our other analyses, we spatially interpolated projections onto a 0.5 × 0.5 m² soil subsidence raster for the study area each year. Internal validation showed that linear interpolation was highly reliable ($R^2 > 0.99$).

Biotic models. Initial distribution of bird territories. We only considered natural habitat that was susceptible to tidal flooding on the islands in our study (that is, not behind a sea wall). Breeding territories were mapped on all islands at least every 5 years using a standard protocol since the 1980s (but not all subareas on an island were censused in the same year)⁶². The number of breeding territories on Dutch island marshes was mostly stable in the 1980s and started declining in the 1990s. The highest densities were thus in the 1980s when the habitat was probably saturated. To model a territory distribution that most resembles the saturated situation at the start year of our model simulation (1986; see next section), we selected the census year that had the highest density and used the territory coordinates as midpoints for each territory (data in ref. 51). Next, we overlaid the territory midpoints with the spatial elevation map and used a Voronoi tessellation algorithm⁶³ to divide all grid cells on the islands among all breeding territories.

Population counts in the same years from our long-term demographic study on island S suggested that the above breeding surveys capture the density distribution of oystercatchers well but that the total number of territories was underestimated. Thus, henceforth, we focus on changes in relative numbers instead of absolute population

numbers, thereby assuming that mapped initial densities are a representative sample of the populations on these islands.

Habitat selection model. Our adaptive habitat selection model was inspired by a previous analysis of settlement patterns in the long-term study population of island S, showing that settlement rates are highest in areas where many young have been successfully raised in the previous 3 years²⁵. The use of such public information⁶⁴ is adaptive, as productive areas reflect territories where environmental conditions are best and can thus be seen as an integrative proxy of territory quality: territories with many offspring tend to be higher territories with less frequent flooding but also can be good in other ways, such as less predation and better access to food. An offspring production over a 3 year period aligns with a long prospecting phase of non-breeders before they decide to settle somewhere (the median age at which oystercatchers typically first settle in a breeding territory is 3 years after sexual maturity⁴⁸).

As flooding events have become much more frequent over time, lower territories increasingly have produced fewer young, making them less attractive for settlement. Others²⁵ further showed that the population on island S has moved their nest elevations up over the decades (5.2 mm yr⁻¹ from 1995 to 2016) and that this change can be explained by relatively low territories having a relatively low probability of becoming re-occupied over time (that is, relatively more of the population breeding in higher parts of the marsh over time, resulting in higher mean nest elevations). The above adaptive habitat selection strategy was implemented such that all vacant territories were ranked according to the number of fledglings they produced over the past 3 years and territories were occupied by non-breeders in that order.

We assumed that territory numbers had reached carrying capacity in the 1980s and thus that, at the start of our simulation, the area was saturated and no more new territories could be created as in ref. 38. Once territories become vacant because of mortality of breeders, they could either remain vacant or become re-occupied by non-breeders recruiting into the population at the start of the breeding season. If there were more non-breeders than vacant territories, all territories were re-occupied. If there were fewer non-breeders than vacant territories, non-breeders used the above adaptive habitat selection strategy to determine which territory to settle in. To assess the buffering contribution of adaptive habitat selection to population change, we compared the above adaptive strategy to a random habitat selection strategy as a null model, in which non-breeders selected a random vacant territory to settle in.

Nest-site selection model. The mean nest elevation of the oystercatcher population on island S has increased over time but this population change over time can be explained by the previously described adaptive habitat selection strategy²⁵. Individual oystercatchers historically have not changed their nest-site selection in terms of nest elevation during their lifetime, in response to environmental cues before the breeding or after experiencing flooding events (no learning³¹).

Despite nest-site selection historically not appearing to be phenotypically plastic to changing flooding risks during a lifetime, nest-site selection is not random. We have almost no records of birds nesting lower than 20 cm above MHT and therefore assumed that birds did not consider this as nesting habitat²⁵. Furthermore, birds typically nest at an above-average elevation within their territory (+6 cm on island S and +11 cm on A^{31,65}, which we averaged for all islands). We included such a nest-site selection rule in our model, as this above-average nest elevation choice is thought to occur because birds prefer vegetation of an intermediate height and such vegetation is linked to relative elevation due to flooding frequency determining vegetation type¹⁸.

Reproductive models. We modelled how annual offspring production (K , number of independent fledglings per territory) depended on the

maximum amount of flooding experienced during a breeding season (X) using a truncated Poisson process, as the maximum clutch size is four eggs in this single-brooded species:

$$K_{y,l} = \text{Truncated Poisson}(e^{\phi_0 + \phi_1(X_{y,l} - \tau) + \phi_2 Q_{y,l} + \phi_3 U_{y,l} + \phi_4 V_{y,l}}) \quad (9)$$

The maximum flooding level X was calculated as the maximum amount of water by which high tides in the 42 days after egg laying exceeded the nest elevation (with $X \geq 0$; the value of 42 is the sum of the 28 day egg incubation period and the 14 day period young chicks are estimated to be at risk of drowning⁴). The level of the highest tides was adjusted for amplification and attenuation of water when travelling over the marsh plain to each specific nest location (using equation (6)). Since eggs can float and chicks can survive a bit of water in their nest, we included a threshold level of $\tau = 0.1$ m above which flooding started to impact survival (with $(X - \tau) \geq 0$) (as also done in ref. 4).

In this reproductive model, we included various extra processes which (indirectly) moderate the flooding risk. Offspring production is not only directly dependent on the timing of egg laying (Q)³⁹ but also indirectly, as earlier nests have lower flooding risk due to the highest tides being typically later in the season⁴. The timing of egg laying for each nest was generated using a random normal process with a mean laying date of 20 May and a standard deviation of 10 days (derived from the long-term studies on S and A).

We also included an effect of the distance to the coast (within 100 m of coastline ($U = 1$) or not ($U = 0$)), as territories adjacent to the coastline can take their chicks directly to the adjacent intertidal foraging grounds and typically produce double the number of offspring per year⁶⁶. Similarly, territories adjacent to a creek also have higher reproductive success due to additional access to food and hiding places for their chicks (within 100 m of a creek ($V = 1$) or not ($V = 0$)). The effects of distance to the coast or creek on reproduction interacts with flooding risk, as these two factors codetermine which territories are most productive and hence which ones will be favoured by habitat selection, which in turn affects future reproductive output. Notably, historically oystercatchers prefer to nest close the intertidal flats and this represents a counter-selection pressure against young birds settling in territories higher up the marsh that are safer flooding-wise but typically further away from the food. Finally, in equation (9) the mean of the Poisson process was also dependent on a constant (ϕ_0) reflecting the mean offspring production in the absence of flooding processes.

We estimated the parameter values, using a truncated Poisson regression model in glmmTMB package⁶⁷ reflecting the structure of equation (9). We applied the model to the long-term nesting data from islands S and A for the years in which nest elevations were measured (see ref. 25 for elevation protocol; S, 1995–2020; A, 2012–2019); for the standardized reproduction monitoring protocol see ref. 66. The estimated parameter values were $\phi_0 = -0.823$, $\phi_1 = -8.22$, $\phi_2 = -0.00767$, $\phi_3 = -1.024$ and $\phi_4 = -0.229$.

Population model. On the basis of an existing population model³⁸ for oystercatchers on island S, we used a postbreeding-census stage-structured stochastic matrix population model⁶⁸ consisting of three subadult age classes (fledglings (F), one-year-olds (O) and two-year-olds (T)) and two adult stage classes (non-breeder (N) and breeder (B)):

$$\begin{pmatrix} n_F \\ n_O \\ n_T \\ n_N \\ n_B \end{pmatrix}_{y+1} = \begin{pmatrix} 0 & 0 & 0 & 0 & J_B K \\ J_F & 0 & 0 & 0 & 0 \\ 0 & J_O & 0 & 0 & 0 \\ 0 & 0 & J_T & J_N(1-E) & 0 \\ 0 & 0 & 0 & J_N E & J_B \end{pmatrix} \begin{pmatrix} n_F \\ n_O \\ n_T \\ n_N \\ n_B \end{pmatrix}_y \quad (10)$$

The birth rate K is a random variable determined by the reproduction model of equation (9) and changes over time due to SLR. The recruitment rate E is determined by the habitat selection model and the number of breeder deaths and is thus density dependent. The stage-dependent survival rates J_{stage} are independent of SLR and modelled as a random Bernoulli process with expectations equal to the mean stage-dependent survival rates on island S ($\bar{J}_F = 0.521$, $\bar{J}_O = 0.749$, $\bar{J}_T = 0.936$, $\bar{J}_N = 0.934$, $\bar{J}_B = 0.928$) (ref. 39). Since the vital rates in our model are modelled as discrete random variables, they ensure integer values for population size and generate demographic stochasticity in addition to environmental stochasticity induced by the variability in sea level.

Model validation. In addition to the model validation already carried out in the original papers that published the submodels described in previous sections, we performed additional model validation of a key output variable of the overall model. We compared the model-projected changes in the number of breeding pairs over the 2000–2020 period with changes in the counted number of breeding pairs over the same period (except for island T, as available count data was too sparse there). The mean annual rate of population change over this period was estimated by taking the exponent of the change in $\log(\text{population size})$ divided by the period duration ($\exp \frac{\log(n_{2020}) - \log(n_{2000})}{2020 - 2000}$). The overall model projections were consistent with observed trends in counts: projected versus observed island A, $-2.5\% \text{ yr}^{-1}$ versus $-2.4\% \text{ yr}^{-1}$; island S, $-2.3\% \text{ yr}^{-1}$ versus $-3.0\% \text{ yr}^{-1}$; and island T, $-3.6\% \text{ yr}^{-1}$ versus no long-term count data available). It should also be noted that these populations would be declining even if there were no SLR (in Fig. 2c the annual rate of population change is negative even when sea levels would not be increasing; $\text{SLR} < 0$), as they also suffer from other problems (high nest predation and low adult survival³⁹).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

No new data were collected, nor field experiments conducted as part of this study. All input data for our models (see Extended Data Table 1 for overview and sources) are described and available via Dryad at <https://doi.org/10.5061/dryad.wm37pvmth> (ref. 51) to facilitate reproduction of all our model results.

Code availability

The R code that can reproduce all results (using R v.4.2.1) is available via Dryad at <https://doi.org/10.5061/dryad.wm37pvmth> (ref. 51). This repository includes a file with all model parameter values, functions which describe the abiotic and biotic submodels from Methods and code to simulate the overall stochastic model with replication and store model results of each scenario considered.

References

- van de Pol, M. et al. A global assessment of the conservation status of the nominate subspecies of Eurasian Oystercatcher *Haematopus ostralegus ostralegus*. *Int. Wader Stud.* **20**, 27–61 (2016).
- van de Pol, M. et al. A silver spoon for a golden future: long-term effects of natal origin on fitness prospects of oystercatchers (*Haematopus ostralegus*). *J. Anim. Ecol.* **75**, 616–626 (2006).
- Ens, B. et al. Chapter Eight—the study of career decisions: oystercatchers as social prisoners. *Adv. Stud. Behav.* **46**, 343–420 (2014).
- Bruinzeel, L. et al. Local dominance and territorial settlement of nonbreeding oystercatchers. *Behaviour* **137**, 473–530 (2000).

50. R Core Team. *R: A Language and Environment for Statistical Computing* (R Foundation for Statistical Computing, 2013).
51. van de Pol, M. Data and code for: Sea level rise causes shorebird population collapse before habitat drowns [Dataset]. *Dryad* <https://doi.org/10.5061/dryad.wm37pvmth> (2024).
52. Haigh, I. et al. Global influences of the 18.61 year nodal cycle and 8.85 year cycle of lunar perigee on high tidal levels. *J. Geophys. Res. Oceans* **116**, C06025 (2011).
53. *Waterhoogte* (Rijkswaterstaat Waterinfo, accessed 10 December 2023); <https://waterinfo.rws.nl/#/kaart/waterhoogte/>
54. Schwalm, C. R. et al. RCP8.5 tracks cumulative CO₂ emissions. *Proc. Natl Acad. Sci. USA* **117**, 19656–19657 (2020).
55. Houston, J. R. & Dean, R. G. Accounting for the nodal tide to improve estimates of sea level acceleration. *J. Coast. Res.* **276**, 801–807 (2011).
56. Baart, F. et al. The effect of the 18.6-year lunar nodal cycle on regional sea-level rise estimates. *J. Coast. Res.* **280**, 511–516 (2012).
57. Stark, J. et al. Observations of tidal and storm surge attenuation in a large tidal marsh: tidal and storm surge attenuation in a marsh. *Limnol. Oceanogr.* **60**, 1371–1381 (2015).
58. AHN Viewer. *Actuele Hoogetekaart Nederland* <https://www.ahn.nl/ahn-viewer> (accessed 12 October 2023).
59. Gombin, J. et al. concaveman: A very fast 2D concave hull algorithm. R package version 1 (2017).
60. Gorelick, N. et al. Google Earth Engine: planetary-scale geospatial analysis for everyone. *Remote Sens. Environ.* **202**, 18–27 (2017).
61. van Wijnen, H. J. & Bakker, J. P. Long-term surface elevation change in salt marshes: a prediction of marsh response to future sea-level rise. *Estuar. Coast. Shelf Sci.* **52**, 381–390 (2001).
62. Dijk, A. J. et al. Introduction of autocluster software in the breeding bird monitoring program. *Limosa* **86**, 94–102 (2013).
63. Schlicht, L. et al. Thiessen polygons as a model for animal territory estimation. *Ibis* **156**, 215–219 (2014).
64. Pärt, T. & Doligez, B. Gathering public information for habitat selection: prospecting birds cue on parental activity. *Proc. R. Soc. Lond. B* **270**, 1809–1813 (2003).
65. Hallmann, C. & Ens, B. in *Monitoring Effecten van Bodemdaling op Ameland-Oost 2005–2010* (ed. de Vlas, J.) 1–147 (Waddenacademie, 2011).
66. Ens, B. J. et al. Territory quality, parental effort and reproductive success of Oystercatchers (*Haematopus ostralegus*). *J. Anim. Ecol.* **61**, 703 (1992).
67. Brooks, M. E. et al. glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. *R J.* **9**, 378 (2017).
68. Caswell, H. *Matrix Population Models* Vol. 1 (Sinauer, 2000).

Acknowledgements

We thank D. le Bars, B. Brinkman, S. Buesink, H. van Dobben, C. Kampichler, H. van der Kolk, M. van Puijenbroek and J. de Vlas for discussion on SLR, marsh sedimentation, deep soil subsidence and shorebirds and Natuurmonumenten and Cooperatie Neerlands Reid for access to the study areas. We are grateful to K. Oosterbeek, J. de Jong, R. Martig and many others for their contribution to fieldwork and to C. Both for long-term support for the oystercatcher research. Funding was provided by the Applied and Engineering Sciences domain of the Netherlands Organisation for Scientific Research (NWO-STW 14638) and cofunding by NAM Gas Exploration, Birdlife Netherlands, Royal Netherlands Air Force and Deltares. This research was cofunded by NAM Gas Exploration and Birdlife Netherlands, two stakeholders that do not always agree on the desirability of mining activities in this nature area. Ecologists working at these organizations provided feedback during half-yearly progress meetings under the guidance of the Netherlands Organisation for Scientific Research but the authors were solely responsible for the research and interpretation of results.

Author contributions

M.v.d.P. conceived the study, with input from L.D.B., B.J.E., M.F., A.M.A., L.B., E.J. and H.d.K. M.v.d.P. constructed the models, performed analyses and led the writing. L.D.B., M.v.d.S., N.H., B.J.E. and M.v.d.P. contributed to field data collection. All authors contributed to paper editing. Funding was applied for by H.d.K., E.J., B.J.E. and M.v.d.P.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41558-024-02051-w>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41558-024-02051-w>.

Correspondence and requests for materials should be addressed to Martijn van de Pol.

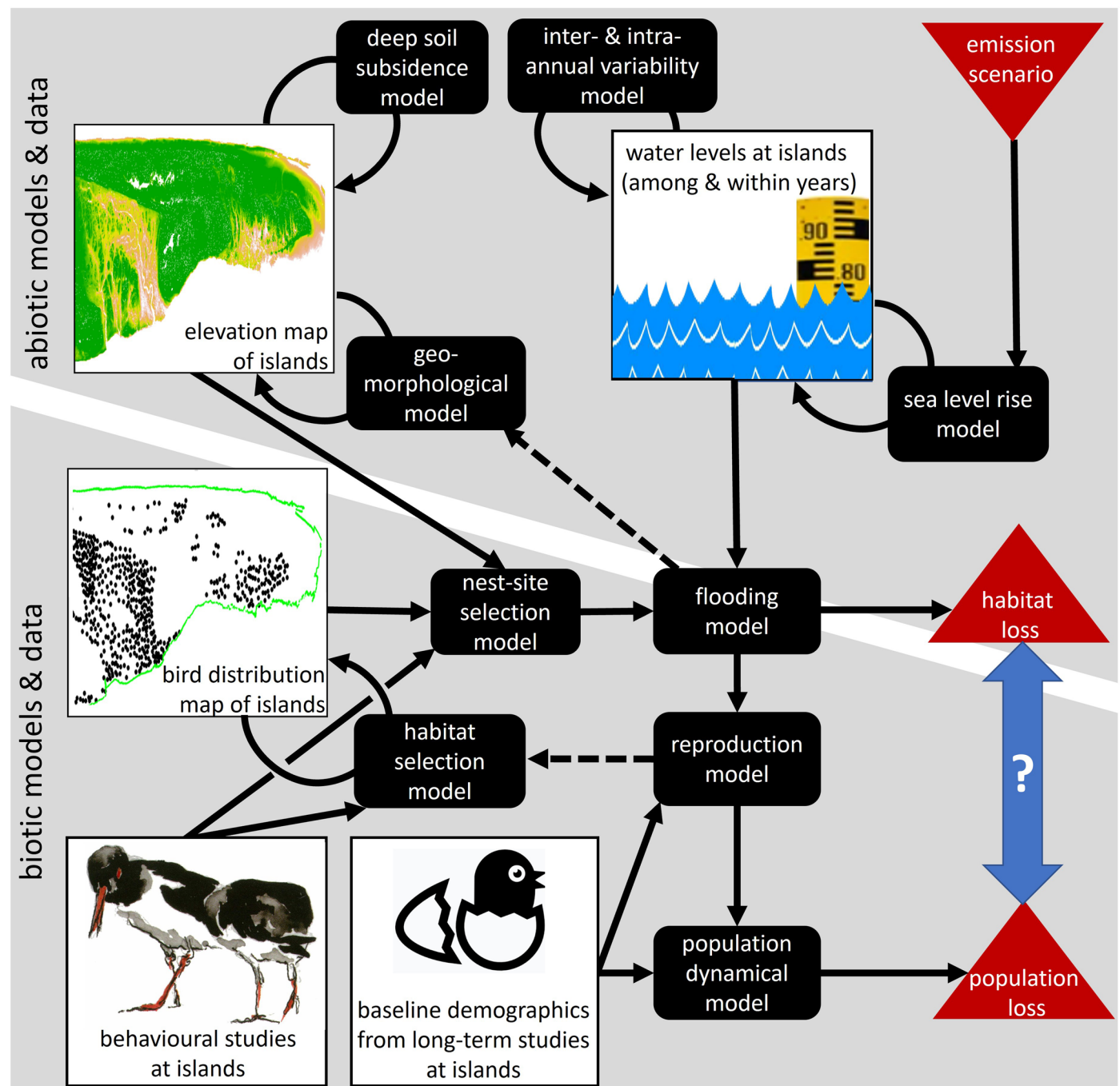
Peer review information *Nature Climate Change* thanks Chris Elphick, Neil Saintilan and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Extended Data Table 1 | Overview of the different submodels and input data sources and how they were related or derived from studies on the three islands

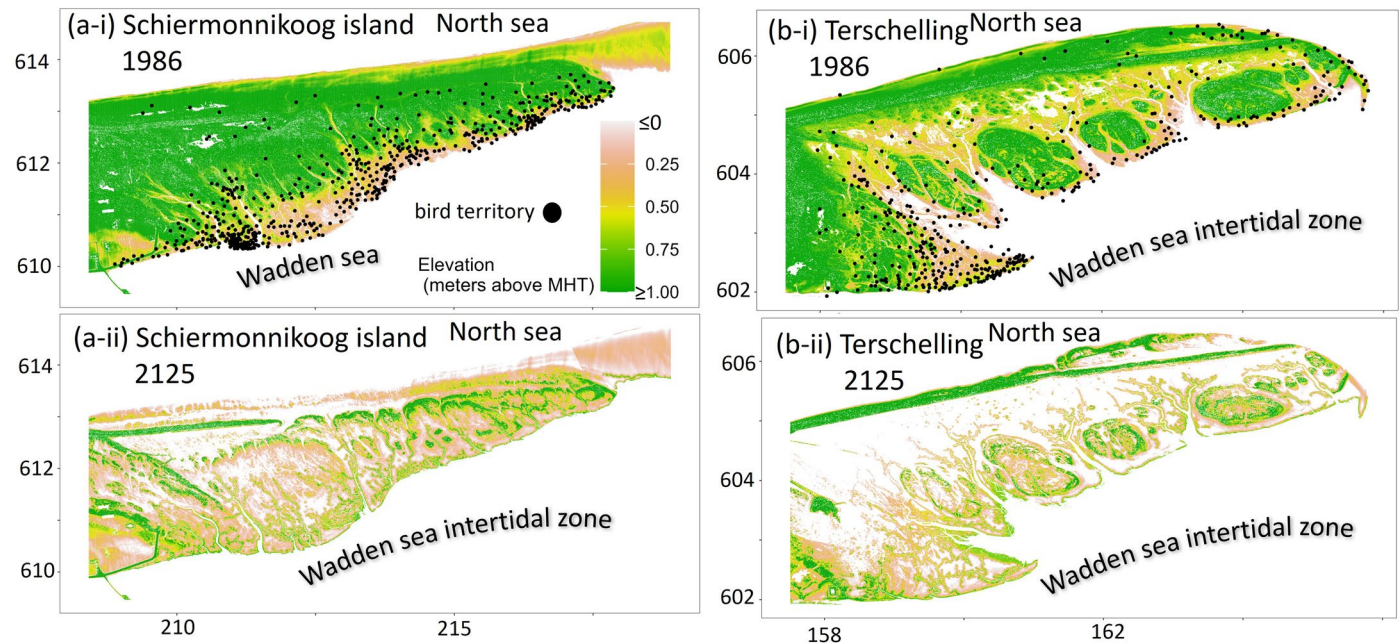
	submodel / input data	island A	island S	island T	source
abiotic models	geomorphological model	same model for all islands, parameterized with data from islands A, S & T			28
	sea level rise model	global IPCC model spatially downscaled for the situation of the Netherlands			27
	intra-and interannual fluctuations in water levels	+	+	+	53
	deep soil subsidence gas mining model	+	not applicable	not applicable	45
	elevation map	+	+	+	58
	flooding model	same model for all islands, parameterized with data from other Dutch marshes			57
biotic models	bird distribution map	+	+	+	62
	habitat selection model	same model for all islands, parameterized with empirical data from island S			25
	nest-site selection model	same model for all islands, parameterized with empirical data from islands S & A			4,31,65
	reproduction model	same model for all islands, parameterized with empirical data from islands S & A			38
	population dynamical model	same model for all islands, parameterized with empirical data from island S			38,39

In the table '+' denotes that the data source or model was specifically collected or parameterized for that specific island.



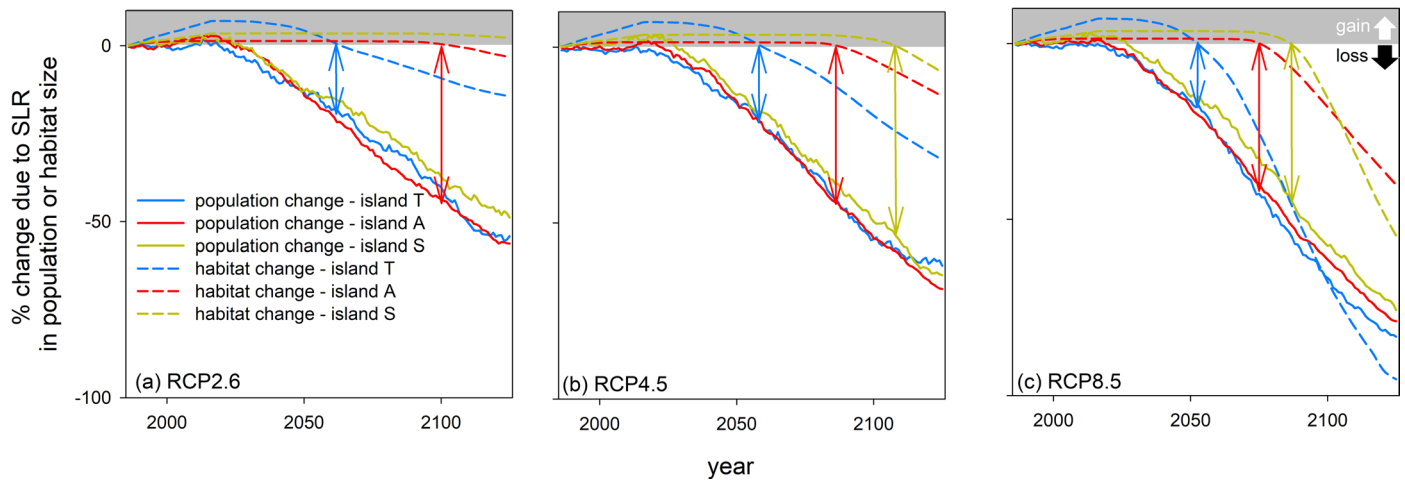
Extended Data Fig. 1 | Schematic overview of models and data sources. A schematic overview of how abiotic and biotic models and data sources are linked to quantify the impact of future emission scenarios on habitat and population size loss of shorebirds. Models (black rectangles) are parameterized and initialized (black arrows) using various data sources (white squares). Model output of a submodel, can feed into another submodel (black arrows) and ultimately be used to quantify the response variable of interest (red triangles; habitat and population size loss). Specifically: greenhouse gas emission scenarios are translated into sea level rise using IPCC scenarios spatially downscaled to the situation of the Netherlands. Sea levels further vary among and within years by drawing from historical intra- and interannual variability in water levels (reflecting weather variability and tidal cycles) and from a lunar nodal cycle model³². Marsh elevation changes over time due to accretion from sedimentation during flooding (geomorphological model; on island A there is additional deep soil subsidence from gas mining), which together with sea

level rise determines habitat loss (that is proportion of habitat submerging below mean high tide). Elevation and sea-level dynamics together determine the relative elevation of the marsh and combined with a flooding model – describing the amplification and attenuation of water along creeks and the marsh plane respectively –, determines the flooding risk at each location. Biotic models for habitat and nest-site selection determine the nesting locations of birds and thereby their flooding pattern. Finally, the flooding level during the breeding season affects the number of offspring produced each year and this birth rate feeds into a population dynamical model that projects changes in population size. Key abiotic and biotic feedback loops (dashed arrows) are (i) higher flooding levels/frequencies leading to higher sediment accretion of marshes and the resulting vertical growth lowers future flooding risk and (ii) lower reproduction due to more frequent flooding leading to fewer birds selecting these (low) territories for settlement, buffering future flooding impact on nests success.



Extended Data Fig. 2 | Elevation and bird distribution maps of the study islands. Elevation maps of islands (a) Schiermonnikoog and (b) Terschelling for the years (i) 1986 and (ii) 2125 under the intermediate RCP4.5 emission scenario. In the map of 1986, the distribution of oystercatchers' territories is shown with a black dot for each territory (based on breeding census data). Elevation is in

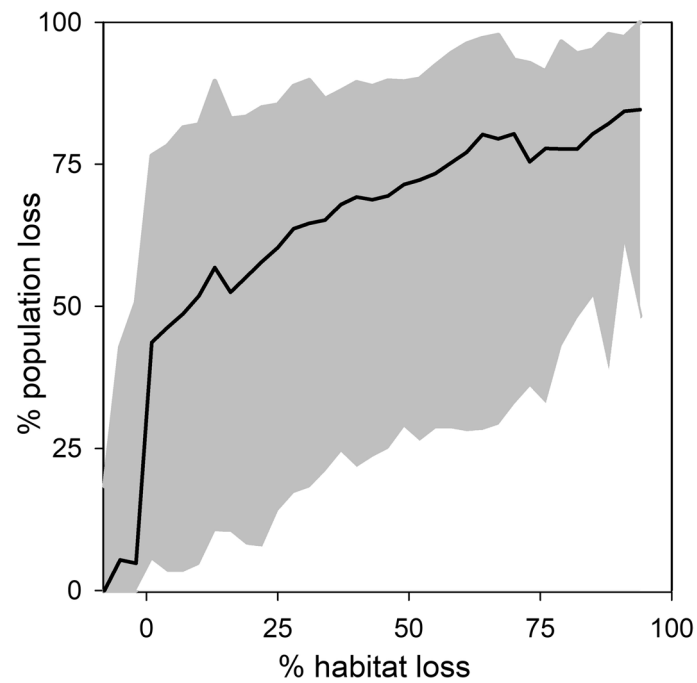
metres above mean high tide (MHT) where the MHT each year is adjusted with the amount of sea level rise (but removed of interannual fluctuations due to weather and the lunar nodal cycle). Coordinates on the axis are from the Dutch national RD grid (kilometre units). For similar maps of island A, see Fig. 1 in the main text.



Extended Data Fig. 3 | Changes in population and habitat size over time for each of the three study islands. Shown are changes under emission scenarios (a) RCP2.6, (b) RCP4.5 and (c) RCP8.5. Population size loss is calculated relative to the population size in the reference of a scenario where SLR would not have accelerated ('no SLR') to determine the impact of additional SLR due to greenhouse gas emissions (for example $\frac{n_{y,RCP4.5} - n_{y,noSLR}}{n_{y,noSLR}}$). The double arrows

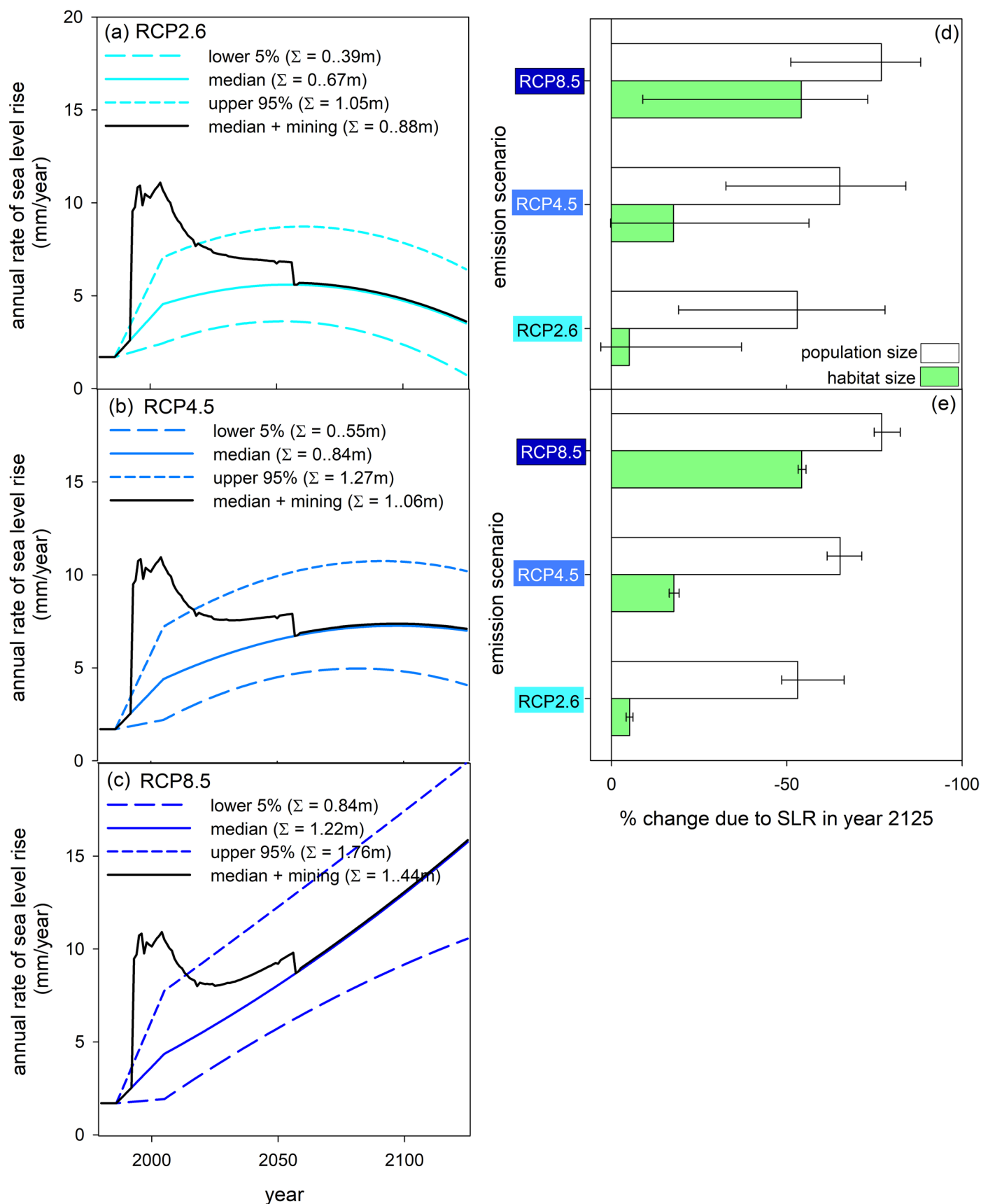
year

highlight the amount of population loss already occurring at the point in time where habitat is first lost. Island T is the lowest island (median elevation of territories and study area is 0.35 m and 0.74 m above MHT, respectively) and therefore loses most habitat, even though island A has the lowest critical rate of SLR (Fig. 2a in main text; the median elevation of territories and study area of island A is 0.67 m and 1.21 m above MHT, respectively).



Extended Data Fig. 4 | The weak association between habitat and population loss at a given point in time across simulations and emission scenarios. For a given amount of habitat loss the median (solid line) and the range (2.5% to 97.5% quantile; grey area) of population loss is plotted, indicating that habitat loss is

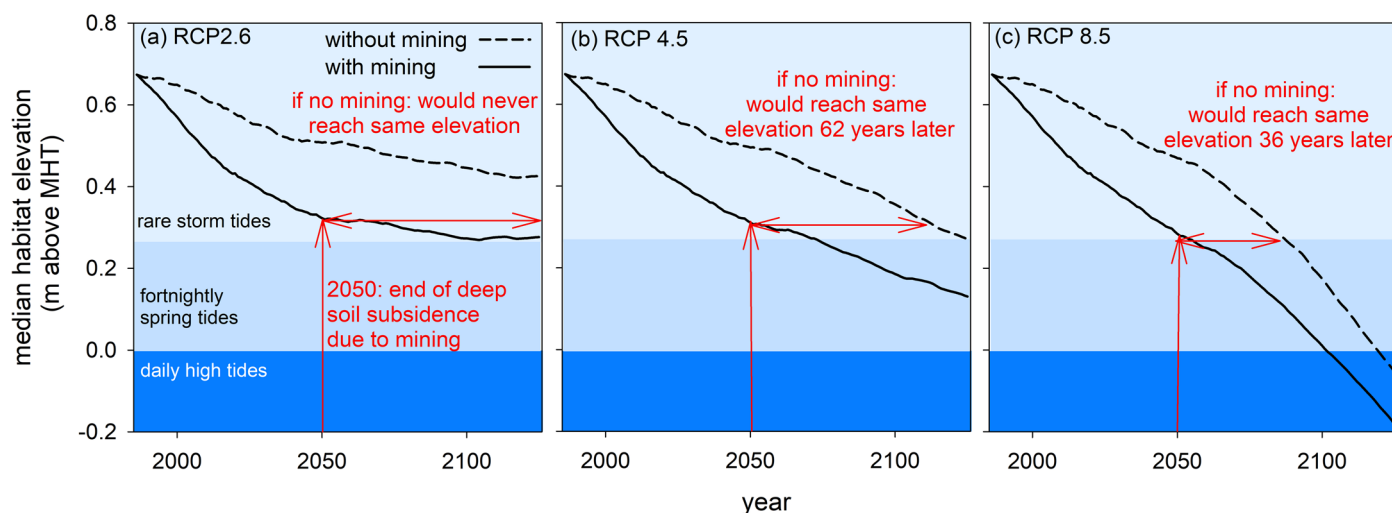
a poor (and nonlinear) predictor of population loss. For example, when habitat loss reached 13%, population loss varied between 10% and 90% across simulations (95% prediction interval).



Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | Uncertainty in model projections. Each greenhouse gas emission scenario has large uncertainty in the rate and amount of sea level rise²⁷, as shown for the **(a)** low RCP2.6, **(b)** intermediate RCP4.5 and **(c)** very high RCP 8.5 scenario. This implies for example that within the low scenario, there is a small possibility that cumulative SLR will substantially exceed the SLR of the median projection for the intermediate scenario (95% RCP2.6 = 1.05 m, 50% RCP4.5 = 0.84 m in 2125). **(d)** This uncertainty in climate projections also leads to large uncertainty in the amount of projected population and habitat loss, as depicted by the error bars reflecting the mean loss for the 5% and 95%

percentiles of climate projection within each emission scenario (n = 100 replicate simulations). For example, within RCP4.5 the median projection is 21% habitat loss in 2125, but there is a small (5%) probability that habitat loss will be 60% on these islands under this scenario (a threefold difference). **(e)** Uncertainty in population and habitat loss projections due to stochastic processes (environmental and demographic stochasticity) is much smaller, as can be seen from the error bars which reflect the 5%-95% confidence intervals around the mean projected population and habitat loss (across the n = 100 replicate model runs for each island and scenario using median climate projections).



Extended Data Fig. 6 | The influence of soil subsidence due to gas mining on marsh elevation over time. Changes in median elevation of the breeding habitat of the oystercatcher population of island A in a scenario with and without gas mining for the (a) low (RCP2.6), (b) intermediate (RCP4.5) and (c) very high (RCP8.5) emission scenarios (See Fig. 3a for corresponding rates of SLR).

Elevations are relative to the sea level of each year. The double arrows show the island elevation in the year 2050 (when deep soil subsidence due to gas mining is projected to stop) and how many years it would have taken to get to this elevation if there had been no mining. The blue zones reflect the different tidal flooding mechanisms acting at different elevation zones.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No software was used to collect data
Data analysis	Analysis done in R v 4.2.1 (packages glmmTMB v.1.9.11 & concaveman v1.1.0) & code is at https://doi.org/10.5061/dryad.wm37pvmth

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All datasets are described in the section Supplementary Files and publicly available at Dryad at <https://doi.org/10.5061/dryad.wm37pvmth>

Human research participants

Policy information about [studies involving human research participants and Sex and Gender in Research](#).

Reporting on sex and gender

Population characteristics

Recruitment

Ethics oversight

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description

Research sample

Sampling strategy

Data collection

Timing and spatial scale

Data exclusions

Reproducibility

Randomization

Blinding

Did the study involve field work? ☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

Methods

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals	Not applicable
Wild animals	This study uses published data Eurasian oystercatchers from various long-term monitoring schemes and previously published studies
Reporting on sex	Both sexes are considered in the empirical estimates of breeding distribution maps and demographic rates. Our population model only consider females, as is common in such population models, as males are generally not a limiting factors and there is no a priori reason to assume that a two-sex model would result in different outcomes. We have previously shown that for island S a two-sex model gives very similar population dynamics as a female-only model (https://doi.org/10.1002/ecy.3615), although this involved a more complex population model.
Field-collected samples	Not applicable
Ethics oversight	The collection of long-term bird monitoring data on islands Schiermonnikoog and Ameland was approved by the Animal Experiments Committee (CCD project AVD25000202010465).

Note that full information on the approval of the study protocol must also be provided in the manuscript.